

Tanbi Ghosh

Curriculum Vitae

Kolkata, West Bengal, India

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LinkedIn Portfolio GitHub

Academic Profile

Final-year Bachelor of Computer Applications student at the Institute of Engineering & Management, Kolkata, with a strong academic record and growing research interest in computational logic, computer vision, intelligent systems, and data-driven problem solving. Maintains a current CGPA of **8.8**, with **94%** in Higher Secondary education. Interested in applying classroom knowledge to socially meaningful technological solutions, particularly in accessibility-focused computer vision applications and intelligent human-computer interaction systems.

Educational Qualifications

- 2023–2026 **Bachelor of Computer Applications (BCA)**, *Institute of Engineering & Management, Kolkata*, CGPA : **8.84**.
- 2023 **Higher Secondary Examination (Class XII)**, *West Bengal Council of Higher Secondary Education (WBCHSE)*, Percentage: **94%**.
- 2021 **Secondary Examination (Class X)**, *West Bengal Board of Secondary Education (WBBSE)*, Percentage: **88.8%**.

Publications

- 2026 **Tanbi Ghosh, Jayanta Paul**, ISL-TextLite: A Lightweight Dual-Stream Deep Learning Framework for Real-Time Indian Sign Language Recognition and Text Rendering, In **ACM Transactions on Knowledge Discovery from Data, Just Accepted, ACM New York, NY (Impact Factor: 5.0)**(Communicated).
- 2026 Jayanta Paul, Abhijit Mitra **Tanbi Ghosh, Jaya Sil**, IndiToon MemeNet: Dual-Path Deep Framework for Person Recognition and Unethical Content Classification Using the Custom Cartoon Meme Dataset, In **Neurocomputing (Impact Factor: 6.0)**(Communicated).

Technical Interests & Expertise

Core Areas	Computer Vision, Machine Learning, Human-Computer Interaction, Accessibility Technologies, Real-Time Intelligent Systems
Programming & Tools	Python, C, C++, Java, HTML, CSS, JavaScript
Technical Foundations	Operating Systems, Machine Learning Foundations, Web Development, Software Development

Projects

- 2025–Present **Real-Time Indian Sign Language Recognition and Text Generation System**, *Computer Vision and Accessibility-Oriented AI Project*.
- Developed a computer-vision-based prototype to recognize Indian Sign Language (ISL) gestures and convert them into English text in real time.
 - Improved gesture detection stability using background noise subtraction and finger-trace optimization for smoother recognition.
 - Built a functional prototype to support communication accessibility for people with hearing and speech impairments.
 - **Tech Stack:** Python, Machine Learning
- 2024–Present **English to Braille Conversion System**, *Computer Vision and Accessibility-Oriented AI Project*.
- Developed a system to convert English text into Braille for visually impaired users.
 - Focused on accuracy and efficient text-to-Braille translation.
 - Aimed to enhance accessibility and support inclusive reading technologies.
 - **Tech Stack:** Python, Machine Learning

Leadership, Editorial & Event Experience

- 2023–2026 **Deputy Secretary**, *Gyan Ke Bytes – Official College News Magazine*.
- Oversaw editorial activities as Chief Content Writer, managed content and publication workflow, and communicated with contributors in the role of Deputy Secretary.
- 2025 **Main Organizer & Student Shark**, *Campus Tank – Entrepreneurship & Business Pitch Event*.
- Organized and led a business-oriented entrepreneurship event, managing planning, coordination, and execution while also participating as a student investor (Shark).
- 2025 **Chess Champion**, *Ashram League – College Tournament*.
- Secured first position in the college-level chess championship, demonstrating strategic thinking and competitive excellence.
- 2023–2026 **Writing & Painting Competitions**, *Multiple College-Level Events*.
- Achieved multiple Champion and Runner-up positions in writing and painting competitions across academic years.

Internships

- 2024 **Web Development Intern**, *ApexPlanet Software Pvt. Ltd.*.
- Worked on web-development-oriented tasks involving front-end technologies and basic application development.
- 2024 **Web Development Intern**, *CodSoft*, January Session.
- Completed structured web development tasks and strengthened basic software development practice.
- 2025 **Journalism Intern**, *NewsTapOne*.
- Gained exposure to structured content writing, information collection, editing, and communication-oriented documentation.
- 2025 **Campus Ambassador**, *SmartED*.
- Contributed to student outreach, communication, coordination, and institutional promotion activities.
- 2025 **MERN Stack Development Intern**, *Euphoria GenX*, Kolkata.
- Gained practical exposure to full-stack web development and project-based software implementation.

Hackathons & Technical Events

- 2023 Participant, **Escape Room Coding Competition**.
- 2025 Participant (2×), **HackArena Hackathon 2025**.
- 2025 Participant, **HACK{0}LUTION 2K25**.

Workshops, Lectures & Academic Participation

- 2025 Participant, **Kolkata Tech Workshop 2025**, organized by Kshitij, IIT Kharagpur.
- 2025 Participant, **Early Brahmi Workshop**, Centre of Excellence for Indian Knowledge Systems (CoE IKS).
- 2025 Participant, **ESI Golden Jubilee Lecture Series: Talk VII**, CoE IKS.

Industry Visit

- 2025 Industry visit to **Grow Ever Steel (India) Pvt. Ltd.**

Skills

- Technical Operating Systems, Python, C, C++, Java, HTML, CSS, JavaScript, Machine Learning Foundations, Web Development, MERN Stack Development
- Non- Problem formulation, prototype development, documentation, data-driven analysis, debugging,
- Technical literature-oriented reading, technical presentation, academic writing, creative writing, content editing, proofreading, Bengali and English writing, team coordination, painting, singing, story writing

Research Motivation

Motivated to gain formal exposure to research methodology, including problem identification, systematic experimentation, evaluation, and scholarly communication. Particularly interested in contributing to research that connects computational techniques with practical human-centered applications.

Interests

Reading storybooks and literary works; listening to music; watching films with appreciation for classical and critically acclaimed cinema.

Languages Known

- Bengali Native proficiency
- English Professional proficiency
- Hindi Professional proficiency

Personal Details

- Date of Birth 14 May 2005
- Nationality Indian